

Digital User Churn Analysis & Dashboard

Business Intelligence Analyst Virtual Internship Program (Bank Muamalat x Rakamin)

Presented by
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Disclaimer

This final project was completed as part of a project-based Business Intelligence Analyst Virtual Internship program, utilizing a structured dataset and an analytical approach supported by data visualization.

BUSINESS CASE

The challenge in this case is to **transform raw transactional data into a structured and integrated dataset that enables meaningful business insights**. This involves **identifying primary keys, defining relationships across tables, and consolidating the data into a master table suitable for analysis**.

The analysis is focused on **evaluating sales performance across key business dimensions** using the following metrics:

- Total overall sales
- Total sales by product category
- Total quantity by product category
- Total sales by city
- Total quantity by city
- Top 5 product categories with the highest sales
- Top 5 product categories with the highest quantity

Through these analyses, the objective is to uncover patterns, trends, and key drivers influencing sales performance from both product and geographic perspectives to **propose data-driven strategies to maintain and improve overall sales performance based on the available transactional data**.

BUSINESS CONTEXT

A Robotics & Drone Education Ecosystem Company leverages its digital platforms to drive user engagement, product adoption, and overall business growth. As digital learning adoption continues to rise, the volume and complexity of user interaction and transactional data have also increased significantly. Understanding purchasing patterns, learner behavior, and product utilization is essential to evaluate sales performance and identify growth opportunities.

To remain competitive and maximize revenue potential, the company needs to effectively transform its data into actionable insights. **Analyzing sales performance across key dimensions, such as product categories (e.g., robot kits, drone kits, blueprints, and training content) and geographic distribution, provides a strong foundation for data-driven decision-making.**

PROBLEM STATEMENT

Business Problem

The company **lacks a structured and comprehensive view of sales performance derived from digital transactions**. Without clear visibility into which products and regions contribute most to revenue, **it becomes challenging to develop effective strategies to sustain or increase sales**.

Key Business Question

As a Business Intelligence Analyst, **what data-driven strategies can be proposed to maintain or increase sales performance based on the available transactional data?**

GOALS

The Main Goals of this analysis is to maintain and improve overall sales performance based on the available transactional data.

By:

- Identify the top-performing product category by sales and orders
- Analyze geographic sales performance
- Provide actionable strategies to improve revenue and growth

PRIMARY KEYS

PRIMARY KEYS of Orders, Customers, Products, Products Category

Query results			
Job information		Results	Visualisation
Row	OrderID	Total_Orders	
1	1098	1	
2	1758	1	
3	404	1	
4	1320	1	

Query results			
Job information		Results	Visualisation
Row	CustomerID	Total_Customers	
1	1480	1	
2	1368	1	
3	260	1	
4	244	1	

Query results			
Job information		Results	Visualisation
Row	ProdNumber	Total_Products	JSON
1	BP106	1	
2	BP104	1	
3	BP108	1	

Query results			
Job information		Results	Visualisation
Row	CategoryID	Total_Category	
1	4	1	
2	1	1	
3	2	1	

QUERY FOR PRIMARY KEYS

Primary Key Orders

```
SELECT OrderID, COUNT(OrderID) as Total_Orders  
FROM `user_churn.Orders`  
GROUP BY OrderID  
ORDER BY Total_Orders  
LIMIT 5;
```

Primary Key Customers

```
SELECT CustomerID, COUNT(CustomerID) as  
Total_Customers  
FROM `user_churn.Customers`  
GROUP BY CustomerID  
ORDER BY Total_Customers  
LIMIT 5;
```

Primary Key Products

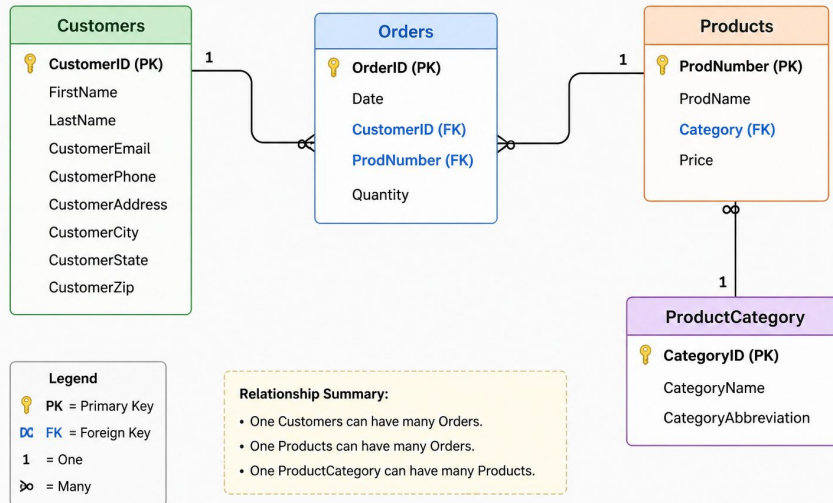
```
SELECT ProdNumber, COUNT(ProdNumber) as  
Total_Products  
FROM `user_churn.Products`  
GROUP BY ProdNumber  
ORDER BY Total_Products  
LIMIT 5;
```

Primary Key Product Category

```
SELECT CategoryID, COUNT(CategoryID) as  
Total_Category  
FROM `user_churn.ProductCategory`  
GROUP BY CategoryID  
ORDER BY Total_Category  
LIMIT 5;
```


ENTITY RELATIONSHIP DIAGRAM (ERD)

ENTITY RELATIONSHIP DIAGRAM



ERD Explanation:

- Customers → Orders**
Type: One-to-Many (1:N)
One customer can have multiple orders, while each order belongs to only one customer.
- Products → Orders**
Type: One-to-Many (1:N)
One product can appear in multiple orders, while each order represents only one product.
- Product Category → Products**
Type: One-to-Many (1:N)
One category can contain multiple products, while each product belongs to only one category.

TABLE MASTER (QUERY JOIN)

Table Master with ordered columns :

- OrderDate (order_date)
- ProductCategoryName (category_name)
- ProductName (product_name)
- ProductPrice (product_price)
- OrderQty (order_qty)
- TotalSales (total_sales)
- CustomerEmail (cust_email)
- CustomerCity (cust_city)

```
SELECT
  o.Date AS order_date,
  pc.CategoryName AS prod_category_name,
  p.ProdName AS product_name,
  p.Price AS product_price,
  o.Quantity AS order_qty,
  (o.Quantity * p.Price) AS total_sales,
  c.CustomerEmail AS cust_email,
  c.CustomerCity AS cust_city
FROM `user_churn.Orders` o
JOIN `user_churn.Customers` c ON o.CustomerID = c.CustomerID
JOIN `user_churn.Products` p ON o.ProdNumber = p.ProdNumber
JOIN `user_churn.ProductCategory` pc ON p.Category = pc.CategoryID
ORDER BY 1 ASC;
```

TABLE MASTER (QUERY RESULT)

Query results								
Job information	Results	Visualisation	JSON	Execution details	Execution graph			
Row	order_date	prod_category_name	product_name	product_price	order_qty	total_sales	cust_email	cust_city
1	2020-01-01	eBooks	Spherical Robots	16.75	5	83.75	lfromonte9@de.vu#mailto:lfo...	Birmingham
2	2020-01-01	Drone Kits	BYOD-220	69.0	1	69.0	edew@nba.com#mailto:edew@...	Honolulu
3	2020-01-01	eBooks	SCARA Robots	19.5	5	97.5	llespercx@com.com#mailto:lle...	Des Moines
4	2020-01-01	eBooks	Polar Robots	23.99	2	47.98	fvaslerqt@comsenz.com#mailt...	Jackson
5	2020-01-01	Robots	RWW-75 Robot	883.0	3	2649.0	tmckernot@tinyurl.com#mailto:...	Katy
6	2020-01-01	Training Videos	Drone Video Techniques	37.99	6	227.94	gstiggersdd@eventbrite.com#mailto:gstiggersdd@eventbrite.com#	Saint Petersburg
7	2020-01-02	Robot Kits	BYOR-2640S	189.0	2	378.0	aguiongo@behance.net#mailto:...	Houston
8	2020-01-02	Training Videos	Drone Video Techniques	37.99	2	75.98	lgatenbyel@quantcast.com#ma...	Des Moines
9	2020-01-02	Robot Kits	BYOR-2640S	189.0	2	378.0	gmirrlees4v@state.tx.us#mailto...	Washington
10	2020-01-02	Blueprints	Ladybug Robot Blueprint	12.0	2	24.0	akingaby78@deviantart.com#m...	West Palm Beach
11	2020-01-02	Drone Kits	BYOD-100	54.0	5	270.0	jzellick84@ustream.tv#mailto:jz...	Washington
12	2020-01-02	Training Videos	Understanding Automation	44.95	1	44.95	ksteershp@ameblo.jp#mailto:k...	San Diego

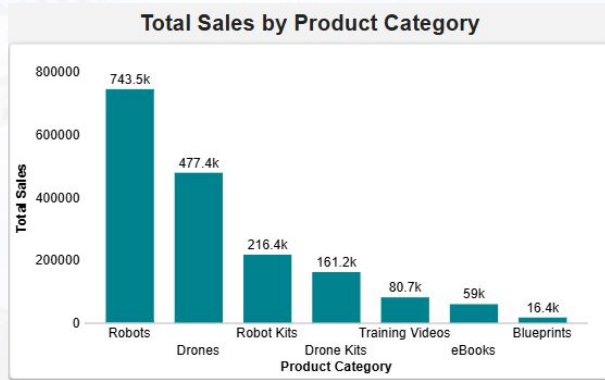
Total Overall Sales and Orders

TOTAL SALES
\$1,754,750.57

TOTAL ORDER QUANTITY
11,654

Total Overall Sales = \$ 1.754.750,57
Total Order Quantity = 11.654

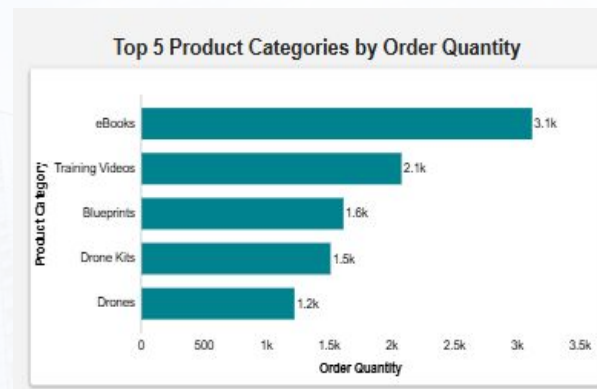
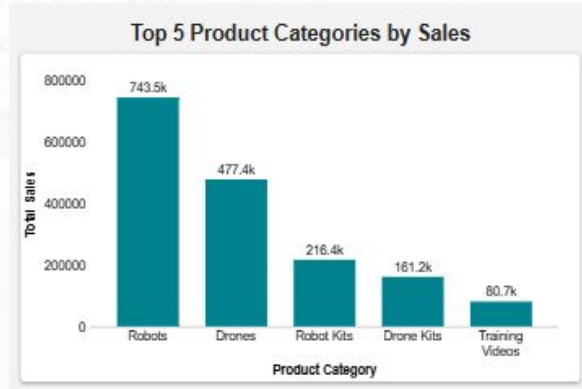
Sales vs Orders Reveal Different Product Winners



- Highest sales: **Robots with 743.5K sales revenue**, followed by **Drones and Robot Kits** in second and third places
- Lowest sales: **Blueprints with 16.4K sales revenue**, followed by **eBooks and Training Videos** in second and third places
- Highest orders: **eBooks with 3.1K order transactions**, followed by **Training Videos and Blueprints** in second and third places
- Lowest orders: **Robot Kits with 1k order transactions**, followed by **Robots and Drones** in second and third places

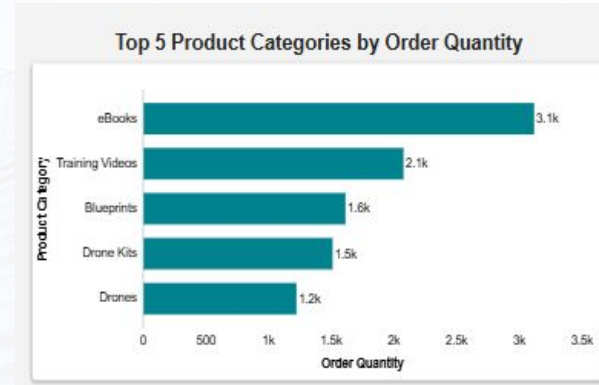
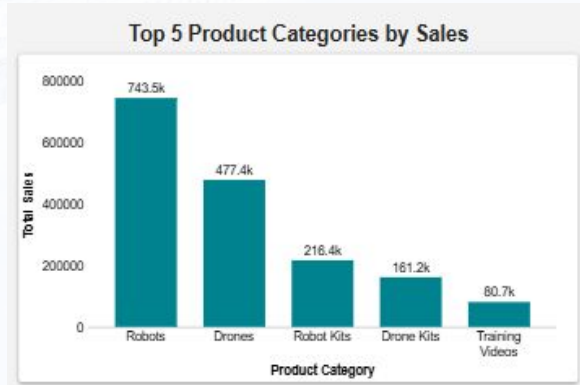
There is a **clear distinction between high-value and high-volume product categories**. This indicates that **revenue is primarily driven by high-value purchases rather than high order quantities**. **High-demand product categories do not necessarily translate into high revenue**. Additionally, **lower adoption of Robot Kits and Robots** may indicate **higher entry barriers or more complexity of the products**.

Top 5 Product Categories: What Drives Revenue vs What Drives Demand?



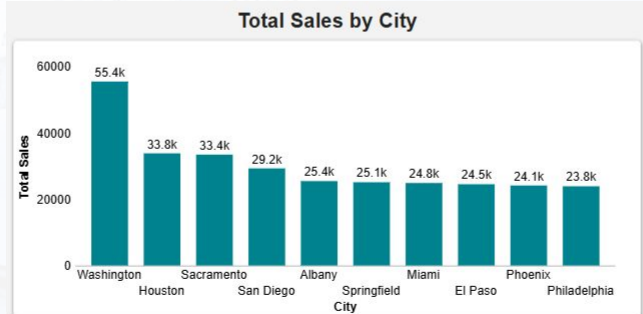
Rank	By Sales	By Quantity
1	Robots (\$743.5k)	eBooks (3.1k)
2	Drones (\$477.4k)	Training Videos (2.1k)
3	Robot Kits (\$216.1k)	Blueprints (1.6k)
4	Drone Kits (\$161.2k)	Drone Kits (1.5k)
5	Training Videos (\$80.7k)	Drones (1.2k)

Top 5 Product Categories: What Drives Revenue vs What Drives Demand?



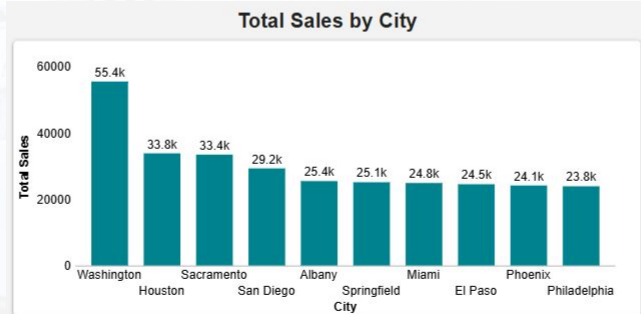
- **High-value vs high-volume:** high-value products (Robots and Drones) drive revenue, while digital contents (eBooks and Training Videos) drive demand.
- **Key Revenue Contributor:** Revenue tends to rely on a few high-value purchases (Robots).
- **Stable Products:** Drone Kits, Drones, and Training Videos are the most stable categories, as they rank among the top 5 product categories across both metrics.

Geographic Performance: Sales & Order Volume by City



- Cities with the highest sales: **Washington with 55.4K sales, Houston with 33.8K sales and Sacramento with 33.4K sales, San Diego with 29.2K sales**
- Cities with the lowest sales: **Philadelphia with 23.8K sales**
- Cities with the highest orders: **Washington with 308 orders, Houston with 249 orders, San Diego with 203 orders.**
- Cities with the lowest orders: **Philadelphia with 139**

Geographic Performance: Sales & Order Volume by City



The geographic analysis shows **a clear performance pattern across cities. Washington leads in both sales (55.4K) and orders (308), followed by Houston (33.8K sales, 249 orders), indicating strong and sustained demand in both regions, further supported by high adoption of tech education.**

Sacramento (33.4K sales) and San Diego (29.2K sales, 203 orders) represent stable mid-tier markets with consistent but lower contributions than the top 2 performing cities.

In contrast, Philadelphia underperforms across both sales (23.8K) and orders (139), indicating weaker market penetration, lower customer engagement, and comparatively lower adoption of tech education.

Overall, **demand is concentrated in Washington and Houston, while Philadelphia remains as a key area for potential growth improvement, depending on customer response and market receptiveness.**

Conclusion

- There is a clear split between **high-value and high-volume products**, where **revenue is driven by high-value complex digital products** (Robots and Drones), while **demand is driven by low-cost digital products** (eBooks and Training Videos).
- **Strong sales performance is not necessarily linked to high order volume**, but **rather to a small number of high-value transactions**.
- **Drone Kits, Drones, and Training Videos** are the most stable categories, as they **perform well in both sales revenue and order volume (in the top 5 in both metrics)**.
- **Washington and Houston consistently lead in both sales and orders**, indicating strong and sustained market demand.
- **Sacramento and San Diego represent mid-tier cities** with stable but **moderate performance across both metrics**, indicating consistent but not yet fully optimized market potential with room for growth through targeted expansion strategies.
- **Philadelphia consistently underperforms in both metrics**, indicating weaker market penetration, lower customer activity, and comparatively lower adoption of tech education.

Actionable Strategy Recommendations

1. **Maximize high-value revenue (Robots & Drones)**
Focus on bundling, upselling, and financing options to improve conversion, strengthen positioning as premium products.
2. **Monetize high-demand products (eBooks & Training Videos)**
Use high-demand products as leads funnel to drive customers toward higher-value products through cross-selling.
3. **Reduce revenue dependency on Robots**
Strengthening **mid-tier products** such as Drone Kits and Drones to avoid over reliance a small number of high-value transactions to improve overall revenue stability.
4. **Scale stable categories (Drone Kits, Drones, Training Videos)**
Expand product variations (bundles, advanced kits, intermediate levels) since they perform well in both revenue and orders.
5. **Improve low-performing products**
Evaluate pricing, complexity, and value proposition of the low-performing products in both sales and orders metrics, simplify offerings and improve usability or education to increase adoption for low-performing products.
6. **Optimize geographic performance**
Strengthen high-performing cities (Washington & Houston) with retention strategies such as loyalty programs and advanced offerings. Drive conversion growth in mid-tier cities (Sacramento & San Diego) through targeted campaigns and optimized product promotion. For the low-performing city (Philadelphia), run awareness campaigns and entry-level promotions (eBooks and Training Videos) to improve market penetration, customer engagement, and adoption.

Link Project & LinkedIn

<https://github.com/grace-nc/bankmuamalat-BI-case-analysis>

<https://datastudio.google.com/reporting/a90dbc1b-22c9-4b38-a2b4-dc39f76ea4cb>

<https://www.linkedin.com/in/grace-natalie-catherine>

Thank You



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